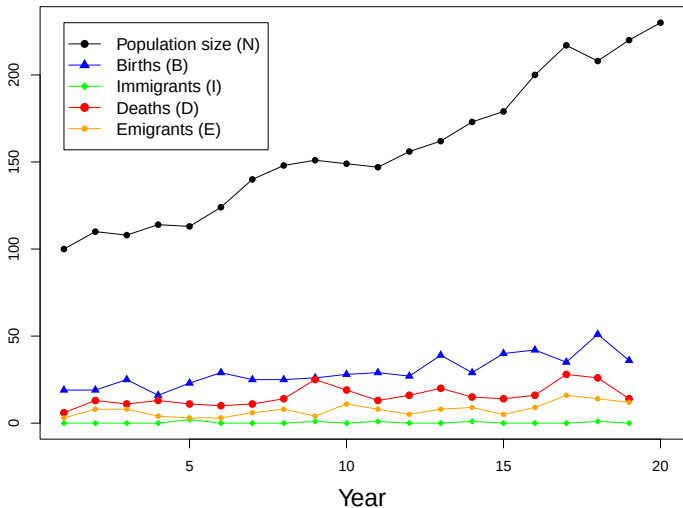


The BIDE model



- 1 DEFINITIONS
- 2 MODELING 101
- 3 BIDE
- 4 ASSIGNMENT

Population dynamics

The study of spatial and temporal variation in population size and structure

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Individuals of the same species occurring in the same geographic region

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Population size and structure

Size: Abundance

Structure: Distribution of individuals among age groups, sexes, habitat patches, etc. . .

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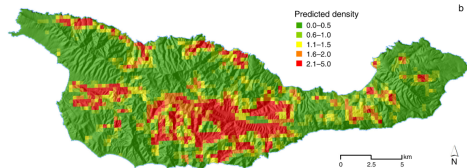
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“All models are wrong, but some are useful.” G.E.P. Box (1987)

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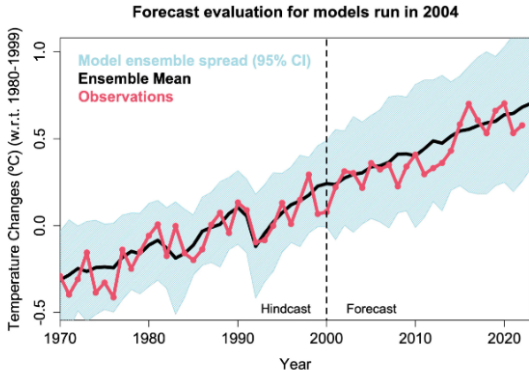
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MODEL VALIDATION

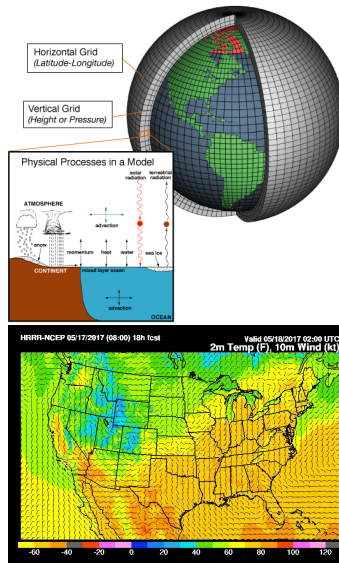
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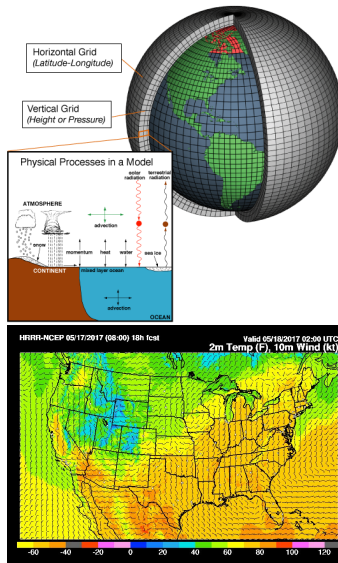
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- Physical
- Graphical
- Mathematical
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$$N_{t+1} = N_t + B_t + I_t - D_t - E_t$$

N_t : population size (state variable) at time t

B_t : births

I_t : immigrants

D_t : deaths

E_t : emigrants

$$N_{t+1} = N_t + B_t + I_t - D_t - E_t$$

As written, this model implies the following:

- B , I , D , and E are not rates, they are the number of events at time t .
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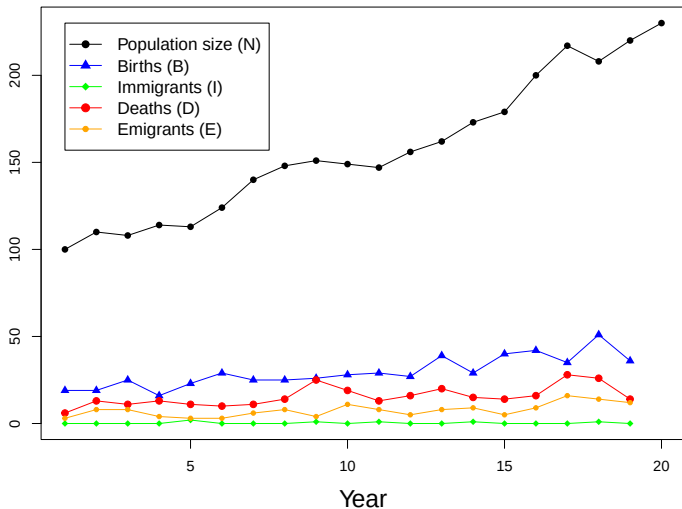
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THE BIDE MODEL



Read the first 3 pages of Chapter 3 in Conroy and Carroll

Expect a quiz next time we meet